CURRICULUM VITAE

David Hansel

Address: INCC, UMR8002, Université Paris Descartes, 45 Rue des Saints-Pères, Paris, France
Telephone: +33 06.07.50.84.03
E-mail: david.hansel@parisdescartes.fr; dhansel0@gmail.com

Education

1986:	Ph.D. in Theoretical Physics; Université Pierre et Marie Curie, Paris.
1982:	Diplôme d'Etudes Approfondies ; Theoretical Physics; Université Pierre et Marie Curie, Paris.
1983:	Military Service
1981:	Licence and Maitrise de Physique ; Université Pierre et Marie Curie, Paris.
1980-1985:	Student at Ecole Normale Supérieure, Paris.
1978-1980:	Student in Classe de Mathématiques Spéciales ; Lycée Louis-Le-Grand, Paris.
1977:	Student in Classe de Mathématiques Supérieures ; Lycée Fénelon, Paris

Academic Positions

Spring 2025:	Visiting professor at Tel Aviv University
March 2025:	Visiting professor at University of California (San Diego)
2020-	Directeur de Recherche (DR1) au CNRS
2015-2021:	Visiting scientist at Janelia Research Campus.
2013-present:	Co-Head (with Carole Levenes) of the research team *Cerebral Dynamics, plasticity, and learning* at Integrative Neuroscience and Cognition Center, Paris Descartes University.
2012-present:	Visiting professor, Alexander Silberman Institute for Life Sciences, The Hebrew University of Jerusalem.
2009-2013:	Head of the research team *Cerebral Dynamics* at Laboratory of Neurophysics and Physiology, Paris Descartes University.
2008-present:	Co-director (with I. Nelken, Hebrew University of Jerusalem, HUJI) of the France-Israel Laboratory of Neuroscience (Laboratoire Européen Associé CNRS-HUJI)
2007:	Morris Belkin Visiting Professorship (4 months; January-April)
2004-2005:	Visiting professor at the Hebrew University of Jerusalem
2003-2020:	Directeur de Recherche (DR2) au CNRS at Laboratory of Neurophysics and Physiology, Paris Descartes University.
2000-2013:	Affiliated researcher at the Interdisciplinary Center for Neural Computation, The Hebrew University, Jerusalem.
1999-2003:	Chargé de Recherche (CR1) au CNRS at Laboratory of Neurophysics and Physiology, Paris Descartes University.

1990-1999: Chargé de Recherche (CR1) au CNRS at Center for Theoretical Physics. Ecole Polytechnique, Palaiseau, France.
1988-1990: The Racah Institute,The Hebrew University of Jerusalem
1986-1990: Chargé de Recherche (CR2, PI) au CNRS at Center for Theoretical Physics Ecole Polytechnique, Palaiseau, France

Other positions

2007-present Chair of *Société internationale de Recherche Emmanuel Levinas* (SIREL, Paris).
2007-2012 Member of the Scientific Committee of the *Fondation France Israel*.
2006-present Life member of the board of the *North American Levinas Society* (NALS, USA).
2004-2007 Chair of the Association pour la Célébration du Centenaire d'Emmanuel Levinas (ACCEL, Paris)

Scientific interests

My interests include modelling of large neuronal networks, the collective dynamics in cortex, the emergence of synchronous activity, mechanisms for persistent activity in executive control (cortex, basal ganglia), as well as selective response of neurons in sensory processing (visual and auditory cortices). I am using analytical calculations as well as numerical simulations of models with various abstraction on levels. The techniques that I am employing are borrowed from theoretical physics (*e.g.,* statistical mechanics, bifurcation theory, theory of extended nonlinear dynamical systems), information theory as well as stochastic processes theory. I am also interested in philosophical questions addressing the role of physico-mathematical approaches in biology, in general, and specifically in brain sciences as well as questions at the interface between science and ethics.

Synergistic Activities

1. *Cross-disciplinary Teaching:* I have trained 8 postdocs, 13 PhD students, 20 undergraduate students to work at the interface between system neuroscience, computational neuroscience and theoretical physics.
2. *The Van Vreeswijk Theoretical Neuroscience Seminar*: Together with C. van Vreeswijk (1962-2022), I have founded the Worldwide Theoretical Neuroscience Seminar (www.wwtns.online). Created as the Worldwide Neuroscience Seminar (WWTNS) in November 2020 and renamed Van Vreeswijk Theoretical Neuroscience Seminar (VVTNS) in homage to Carl van Vreeswijk in Memoriam (April 20, 2022), its aim is to be a platform to exchange ideas among theoreticians. Speakers have the occasion to talk about theoretical aspects of their work which cannot be discussed in a setting where most of the audience consists of experimentalists. The seminars, held on Wednesdays at 11 am ET, are 45-50 min long followed by a discussion. The talks are recorded with authorization of the speaker and are available to everybody on the WWTNS YouTube channel. 2024/2025 is the fourth season of the seminar. Since the first talk held in November 2020, more than 150 speakers have been presenting at VVTNS.
3. *The NeuroBridges summer school:* Since 2017, I have been co-organizing, together with Yonatan Loewenstein (The Hebrew University of Jerusalem) and Ahmed El-Hady (now at the Max Planck in Konstanz, Germany) the summer school NeuroBridges. It takes place at Le Centre de Conferences Internationales de Cluny, France (CCIC), in general in September.

The school provides an overview of theoretical and experimental frameworks used to study a variety of behavior, in particular decision-making processes. It covers different model systems from rodents to humans and a plethora of mechanisms. Another objective of NeuroBridges is to bring together Mediterranean and Middle Eastern scientists, to promote scientific cooperation between young researchers from these countries. The organizers of NeuroBridges are convinced that such scientific collaborations can lead to personal relations, which eventually may alleviate the political distress in the Middle East.
The school is intended for graduate students and postdocs, primarily, but not only, from the Middle East and the Mediterranean region, either working in their home countries or abroad. Applicants should have some background in related fields in neuroscience or cognitive psychology.

4. *Organization of workshops and conferences in neuroscience:* Since the beginning of my career, I have been organizing more than 40 international workshops and conferences, at the interface between theoretical and experimental neuroscience including 36 since 1999.
5. *Organization of workshops at the interface of brain sciences and philosophy:* I have co-organized such workshop in 2017 and 2019 in collaboration with the philosopher Flora Bastiani (MCF, Université de Toulouse Jean-Jaurès), see list below).
6. *Commemoration of the legacy of Carl van Vreeswijk (1962-2022):* I coordinated the organization of the international conference, *Brains at the crossroads of theory and experiments- A worldwide homage to Carl van Vreeswijk (1962-2022),* June 5-8, 2023, Jerusalem. The conference brought together 50 leading neuroscientists whose research is at the crossroads of theory and experiment. They addressed advances and perspectives in our understanding of the brain at our epoch of an explosive growth in experimental tools allowing probing and manipulating brain circuits and behaviors at unprecedented scale and detail.
7. *Faculty member in schools.*
   1999: EU Advanced Course in Computational Neuroscience (Trieste, Italy).
   2003: Les Houches; LXXX Summer School (France).
   2008: European Course in Computational Neuroscience. (Freiburg, RFA).
   2010: Summer School in Computational Neuroscience (Valparaiso, Chile).
   2014: Laconeu Summer School in Computational Neuroscience (Valparaiso, Chile).
   2017-: NeuroBridges Summer School in Computational Neuroscience (Cluny, France)
8. 2022-2023: *Online Seminar series ADDICTS (20in French) for non-specialists*.
9. In 2007, I founded the Société internationale de recherche Emmanuel Levinas (Emmanuel Levinas International Society). Since 2010, SIREL has been holding an annual workshop. It takes place annually during the first week of July (2010-2021 in Toulouse; 2022- at the Emmanuel Levinas Center in Kaunas, Lithuania).
10. I am life member of the board of the North American Levinas Society (USA).

<u>Publications (last ten years)</u>

1. R. Darshan, A. Leblois & D. Hansel, *Interference and Shaping in Sensorimotor Adaptations with Rewards*, *PLOS Comput. Biol. 10: e1003377, 2014*
2. O. Harish & D. Hansel, *Asynchronous rate chaos in neuronal circuits*, PLOS Computational Biology, 10.1371/journal.pcbi.1004266, 2015.

3. A. Barri, Y. Wang, D. Hansel, & G. Mongillo, *Quantifying repetitive transmission at chemical synapses: A generative-model Approach.,* eNeuro. 2016 May 13;3(2). pii: ENEURO.0113-15.2016. doi: 10.1523/ENEURO.0113-15.2016. eCollection 2016 Mar-Apr.
4. T. Arakaki, A. Leblois, S. Charpier & D. Hansel, *The role of striatal feedforward inhibition is the maintenance for absence seizures*, J Neurosci. 36:9618-32, 2016.
5. R Engelken, F Farkhooi, D Hansel*, C van Vreeswijk*, F Wolf*, *Joint senior authorship, Comment *on "Two types of asynchronous activity in networks of excitatory and inhibitory spiking neurons"* bioRxiv 017798; doi: https://doi.org/10.1101/017798
6. R. Engelken, F. Farkhooi, D. Hansel, C. van Vreeswijk, F. Wolf*, A reanalysis of "Two types of asynchronous activity in networks of excitatory and inhibitory spiking neurons",* F1000Res, 5:2043. Doi: 10.12688/f1000research.9144.1.eCollection 2016.
7. D. Darshan, W.E. Wood, S. Peters, A. Leblois, D. Hansel, *A canonical neural mechanism for behavioral variability,* Nature Comm. 8:15415. doi: 10.1038/ncomms15415, 2017.
8. JJ Pattadkal, G Mato, C van Vreeswijk, NJ Priebe, D Hansel, *Emergent Orientation Selectivity from Random Networks in Mouse Visual Cortex*. Cell Rep. 2018 Aug 21;24(8): 2042-2050.e6. doi: 10.1016/j.celrep.2018.07.054.
9. R Darshan, C van Vreeswijk, and D Hansel, *Strength of Correlations in Strongly Recurrent Neuronal Networks* Phys. Rev. X, 8, 031072 2018.
10. S. Rao-Balappa, C. van Vreeswijk and D. Hansel, *Dynamics and orientation selectivity in a cortical model of rodent V1 with excess bidirectional connections,* Scientific Rep., 9, 1–10, 2019.
11. L. Lebovich, R Darshan, Y. Lavi, D.Hansel* and Y. Loewenstein*, *Idiosyncratic choice bias in decision tasks naturally emerges from neuronal network dynamics*, arXiv: 1803.07193, bioRxiv 284877; doi: https://doi.org/10.1101/28487,; Nature Human Behavior, doi.org/10.1038/s41562-019-0682-7, 2019.
12. N. Li, S. Chen, Z. V. Guo, H. Chen, Y. Huo, H. K. Inagaki, C. Davis, D. Hansel, C. Guo, and K. Svoboda, Spatiotemporal *limits of optogenetic manipulations in cortical circuits.* doi.org/10.7554/eLife.48622, *eLife* 8: e48622
13. A. Mahrach, G. Chen, N. Li, C. van Vreeswijk and D. Hansel, *Mechanisms underlying the response of mouse cortical networks to optogenetic manipulation,* eLife, DOI: 10.7554/eLife.49967, 2020;9:e49967
14. A. Suarez-Perez, O. Harish & D. Hansel, *Network mechanisms of working memory: the role of neuronal nonlinearities. 2021, arXiv* :2112.0509.
15. L. Lebovich, L. Kaplan, D. Hansel and Y. Loewenstein, *Idiosyncratic choice bias and feedback-induced bias differ in their long-term dynamics*, 2025 May 17;3(1):79. doi: 10.1038/s44271-025-00263-0. PMID: 40379772; PMCID: PMC12084652.
16. J. Pattadkal, R. O'Shea, D. Hansel, T. Taillefumier, D. Brager and N. J. Priebe, *Synchrony underlies irregular neocortical spiking,* Neuron. 2025 Dec 17:S0896-6273(25)00852-9. doi: 10.1016/j.neuron.2025.11.005. PMID: 39464165; PMCID: PMC11507790.C.
17. A. Leblois, T. Boraud and D. Hansel. *Reward-driven adaptation of movements requires strong recurrent basal ganglia-cortical loops.* Proc Natl Acad Sci U S A. 2025 Dec 16;122(50):e2515994122. doi: 10.1073/pnas.2515994122. Epub 2025 Dec 11. PMID: 41379994; PMCID: PMC12718356.
18. L. Jennings, D. Hansel, N. Priebe. *The fluctuation-based regime of thalamocortical circuitry* bioRxiv 2025.11.20.689511; doi: https://doi.org/10.1101/2025.11.20.689511, in revision

B. Submitted or in preparation

1. C. van Vreeswijk, G. Mato, D. Barbera, J.J. Pattadkal, N.J. Priebe* and D. Hansel*. The computational power of randomness: *An analysis of selectivity in the thalamic input to mouse visual cortex*, submitted.
2. A. Berrou, M. Harpaz, M Jankowski; D. Hansel, and I. Nelken. *The anatomical mixing of responsive and non-responsive neurons in the ascending auditory pathway,* in preparation
3. A. Berrou, M. Harpaz, M Jankowski, D. Hansel, and I. Nelken. *The emergence of heterogeneous neural responses in the ascending auditory pathway,* in preparation
4. A. Berrou, R. Darshan, G. Mato, D. Hansel and C. van Vreeswijk. *Theory of orientation selectivity in random network models of the rodent primary visual cortex*, in preparation.

<u>Invited talks (last ten years)</u>

1. January 20-24, 2014, LACONEU Summer school, Valparaiso, Chile. Talk 1: *The balanced state: the standard model and beyond.* Talk 2: *The mechanism of orientation selectivity in rodents.*
2. February 11-14, 2014, Fifth France-Israel Binational Conference in Neuroscience, Sde Boker, Israel.
3. February 26-27, 2014, Workshop on cortical dynamics: from the architecture to functions, (University of Kyoto, Japan). *The balanced state: the standard model and beyond.*
4. June 10-13, 2014, Sino-French Conference in Computational Neuroscience (Jiao Tong University, Shanghai, China). *Chaotic Rate Dynamics in Large Neuronal Networks.*
5. October 21-23, 2014, Animal vocalizations: Songs in Jerusalem (Jerusalem, Israel). *A network mechanism accounting for universality in babbling vocalization.*
6. December 1-3, 2014, NEUROFrames: pre-kick off meeting of the International Research Network in Neurosciences France- South America (Buenos Aires, Argentina).
7. February 9-12, 2015, Workshop on Computational Neuroscience, (NYU-Abu Dhabi, EAU). *A network mechanism accounting for universality in babbling vocalization.*
8. May 26-27, 2015, Annual retreat of the Department of Neuroscience of Pasteur Institute. (Paris). *The Balanced State: From Physics to Function.*
9. June 8-10, 2015, Meeting in honor of David McLaughlin, on his seventieth birthday (NYU Shanghai, Shanghai, China*). Central network mechanisms of babbling behavior.*
10. July 22-23, 2015, Workshop on oscillations, 2015 CNS Conference (Prague, Czech Republic, replaced by T. Arakaki).
11. July 27-31, 2015, Dynamics of Coupled Oscillators: 40 years of the Kuramoto Model (DYCOSC15), (Max Planck Institute for the Physics of Complex Systems, Dresden, Germany, canceled).
12. 2-12, September 2015, 13th Advanced Summer Course on Computational Neuroscience, Göttingen, *Rate chaos in neuronal networks.*
13. February 9, 2016*,* COSYNE2016, Snowbird, USA. Workshop: Coding, correlations, and the dimensionality of neural activity. *Random feedforward connections account for orientation selectivity of the thalamic input in rodent visual cortex.*
14. March 1, 2016, COSYNE2016, Snowbird, USA. Workshop: Recurrent spiking neural networks - dynamics, learning, computation. *On spikes, rates and chaos.*

15. May 19-20, 2016, The Early Visual System: comprehensive and data-driven modeling, European Institute for Theoretical Neuroscience, Paris, France. *Theory of orientation selectivity in mouse primary visual cortex.*
16. June 24-25, 2016, Neuropsychiatric disorders -from basic systems to clinical applications: Czech and Slovak Neuroscience Alumni Workshop (Institute of Physiology, the Academy of Sciences Campus, Prague, Czech Republic). *Theory of orientation selectivity without map.*
17. July 10-15, 2016, Sixth French-Israel Neurosciences Conference, (CIRM, Marseille, France). *Balanced or not balanced, what is the question?*
18. December 6-8, 2016, Finding the right balance, (Mitzpe Ramon, Israel). *Functional selectivity in random cortical networks: the role of recurrent connections.*
19. May 16-18, 2017, Stochastic Dynamics Out of Equilibrium (Institut Henri Poincaré, Paris). *The correlated cortex.*
20. June 12-15, 2017*,* 6th Conference of the Mediterranean Neuroscience Society (MNS, St Julian, Malta).
21. July 12-13, 2017, 1st A*Midex CompNeuro Days, Workshop on Computational Neuroscience in Marseille (France). *Mice are not Cats: An (almost) exact proof* (cancelled).
22. December *4*-6, 2017, Computational properties of basal ganglia neurons international research workshop of the Israel Science Foundation UK-Israel SYNERGY meeting. (Ramat Gan, Israel). *Recurrent dynamics and competition in basal ganglia.*
23. July 20, 2018, Balance or not balance: what is the question? Colloquium of the Institute of Natural Sciences, (Shanghai Jiao Tong University, Shanghai, China).
24. July 23-27, 2018, International Conference on Applied Mathematics and Computational Neuroscience, In Memory of David Cai *(*Shanghai Jiao Tong University, Shanghai, China).
25. August 6-9, 2018, SIAM Conference on the Life Sciences on Information Processing in Neuronal Networks. *Mice are not Cats: An (almost) exact proof. (*Minneapolis, USA).
26. August-September 2018, *BRAIN*18*:* Recording, analyzing, manipulating, and modeling whole brain activity, (UC Santa Barbara, Kavli Institute for Theoretical Physics, Santa Barbara USA). *Emergent of orientation selectivity from random networks in mouse primary visual cortex.*
27. October 12, 2018, INC Day, *Paris Descartes University (Paris). Mice are not cats: An (almost) exact proof.*
28. April 28-May 2, 2019, IBAG2019, Biarritz (France). *Dynamic properties of the cortex-BG loop.*
29. September 2-6, 2019, French German WE-Heraeus-Seminar / Novel Physics in Living Systems? (Roscoff, France) *Mechanisms underlying the response of mouse cortical networks to optogenetic manipulation.*
30. November 3-6, 2019, The Operating Regime of Neural Circuits as a Determinant for Computations (HHMI Janelia Research Campus, Ashburn, USA), *Theory of orientation selectivity in the rodent primary visual cortex.*
31. January 27-31, 2020, Simons Workshop: Physics of neural circuits and network dynamics. (Stony Brook University, USA), *The power of randomness or why mice are not cats.*
32. April 30, 2020, ELSC Weekly Seminar. *Theory of orientation selectivity in mouse primary visual cortex.* The Hebrew University of Jerusalem, Israel. Postponed to 2021.
33. May 11-13, 2020, China-France Workshop: Multiscale dynamics and structure in the brain: Experimental, Computational and Theoretical Challenges *The power of randomness in early vision.* Online.

34. August 23-27, 2021, Dynamics Days Europe 2021, Nice, France *A function for the balanced state: emergence of selective neural responses*
35. January 17-23, 2022, 11th computational neuroscience winter school in Shanghai Jiao Tong University (China), *Theory of orientation selectivity in mouse primary visual cortex.* Online
36. 13-14, September 2022, Bernstein Conference on Computational Neuroscience, Symposium "Major transition in cortical circuit evolution" (Berlin). *The statistics of the thalamo-cortical excitation in mouse V1 layer 4.*
37. April 23-29, 2023, Workshop on Mathematical foundations of biological organization, Mathematisches Forschunsinstiturt Oberwolfach (RFA). The *thalamo-cortical projections in mouse V1: theory vs. experiments. (Declined for personal reasons)*
38. March 8, 2024, Hong-Kong University of Science and Technology, Hong-Kong. *The power of randomness.*
39. March 19, 2024, Jia tong University, Shanghai, China. *Emergent feature selectivity from random networks in rodent visual cortex.*
40. March 21, 2024: New York University, Shanghai, China. *Thalamocortical receptive fields in rodent: Theory vs. experiments.*
41. April 12, 2024, Kleinfeld lab, UCSD, online. *Receptive fields of thalamo-cortical excitation in rodent visual cortex (part1).*
42. May 5, 2024, Kleinfeld lab, UCSD, online. *Receptive fields of thalamo-cortical excitation in rodent visual cortex (part2).*
43. August 4, 2024, Emilio Segre Distinguished Lectures in Physics. Tel Aviv University, Israel. *Physics of the Brains: Between Diversity and Universality.* Postponed.
44. June 15, 2025, Emilio Segre Distinguished Lectures in Physics. Tel Aviv University, Israel. *Physics of the Brains: Between Diversity and Universality.* Postponed because of the '12 days war'.
45. July 1-3, 2026, Symposium in honor of Yves Frégnac. *Feature selectivity in random networks: Theory, prediction and experiment.*

<u>Grants (last ten years)</u>

2014 PEPS-PTI (Interdisciplinary programs, CNRS). *Physics of the balance state in cortex.* 4 k€.

2014-2017 ANR (generic). *How the balance of excitation and inhibition shapes orientation selectivity and center-surround interactions in the superficial layers of area V1: a combined experimental and modeling project* (with Lionel Nowak and Frederic Chavane). 512 k€.

2015-2018 ANR-US-French Collaboration (CRCNS/NIH/NSF/ANR). *The balance of excitation and inhibition in sensory cortex* (with Nicholas Priebe and Carl van Vreeswijk). 309 k€ for my team.

2015-2018 Human Frontier Science Program. *Deciphering brain oxytocin circuits controlling social behavior*, with Shlomo Wagner, (Haifa), Valery Grinvich (Heidelberg) and Joseph Buxbaum (New York). 350 k$ for my team.

2017-2020 ANR-US-French-Israel Collaboration (CRCNS/NSF/ANR/BSF). *Processing of complex sounds: Cortical Network Mechanisms and Computations.* With Israel Nelken (Jerusalem, Israel) and Tatyana Sharpee (San Diego, USA). 380 k€ for my team.

2017-2022 International Research Network (IRN, former GDRI): *France-Israel Center for Neural Computation.* Co-directed with Israel Nelken (Jerusalem., Israel). 40 k€/year (approx).

2015-2020 HHMI-Janelia Farm. *Visiting Scientists Program*. 15 k$/year (approx).

2023-2027 ANR-US-French Collaboration (CRCNS/ANR/DOE). *A combined computational and experimental investigation of the cellular and network basis of visual recognition*, with Mark Bear (MIT), 434 k€ for my team.

2024-2028 ANR-US-French-Israel Collaboration (CRCNS/ANR/BSF/NIH). *Thalamocortical vs. recurrent connectivity in active sensation of vibrissa touch*, with David Golomb (Ben Gurion University, Israel) and David Kleinfeld (University of California, San Diego, USA), 330 k€ (approx.) for my team.

<u>Organization of workshops and schools (last ten years)</u>

1. December 14, 2009: The 7th Paris Area Computational Neuroscience Day. *Synaptic plasticity: From receptors to networks* (with N. Brunel and D. Holcman, Paris).
2. February 14-18, 2010: Third France-Israel binational conference in Neuroscience, neurology and psychiatry (with G. Richter Levin, M. Maroun and A. Desmedt, Haifa).
3. March 22-24, 2010: Workshop. *Working Memory and Executive control* (with M. Tsodyks, Rehovot, Israel).
4. March 8-9, 2011: International symposium: *Working memory and decision making* (with G. Mongillo, Paris).
5. March 23-25, 2011: USTC-CNRS Joint Symposium. *Synaptic Plasticity and Network Dynamics in the Central Nervous System* (with G. Bi, Heifei, China).
6. December 14-15, 2011: International workshop. *Physiological mechanisms underlying of song learning and production* (with A. Leblois and D. Perkel, Université Paris Descartes).
7. March 25-29, 2012: Fourth France-Israel binational conference in Neuroscience, neurology and psychiatry (with D. Shulz, I. Nelken and A. Desmedt, Aussois, France).
8. December 3-6, 2012: French Chinese Joint Symposium. *Synaptic Plasticity and Network Dynamics in the Central Nervous System* (with G. Bi, Université Paris Descartes).
9. June 10-14, 2013: International conference. *Dynamics of cortical and cortical-subcortical circuits: Interaction between theory and experiments* (with D. Golomb, M. Chklovski, K. Svoboda and J. Dudman, Janelia Farm Research Campus, USA).
10. January 8, 2014: Symposia Series Open questions in Neuroscience- Symposium #1: *Balance of excitation and inhibition in sensory cortex* (coordinator, with F. Chavane and L. Nowak, Université Paris Descartes).
11. March 11, 2014: Symposia Series Open questions in Neuroscience- Symposium #2: *Dynamics of large networks, random matrices and neuroscience* (coordinator, with G. Mongillo and C. van Vreeswijk).
12. May 13, 2014, Symposia Series Open questions in Neuroscience- Symposium #3: New Insights in the Mechanism of Absence Epilepsy (with A. Leblois, Université Paris Descartes).
13. December 9, 2014, Symposia Series Open questions in Neuroscience- Symposium #4: *Plasticity and memory: beyond synapses* (with C. Levenes and G. Mongillo, Université Paris Descartes).
14. July 11-15, 2016, Sixth France-Israel binational conference in Neuroscience, neurology and psychiatry (with F. Chavane and I. Nelken, Marseille, France).

15. Sept 4-14, 2017, NeuroBridges Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA).
16. July 4-6, 2017, International conference at the interface of neuroscience and ethics. *Life between Ethics and Science*, Toulouse, France (Université de Toulouse Jean-Jaurès, with Flora Bastiani).
17. October 1-12, 2018: NeuroBridge2018 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University and A. El Hady, Princeton, USA).
18. July 16-17, 2019: CNS2019 workshop on New Perspectives on Cortical Dynamics (Barcelona, Spain, with Ni. Priebe and C. Van Vreeswijk).
19. July 1-3, 2019, International conference at the interface of neuroscience and ethics. *Vie et relation* (Université de Toulouse Jean-Jaurès, Toulouse, France, with Flora Bastiani).
20. September 8-19, 2019, NeuroBridge2019 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA)
21. September 6-17, 2020, NeuroBridge2020 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA). Cancelled
22. May 11-13, 2020, China-France Workshop: *Multiscale dynamics and structure in the brain*. (Online, With G. Bi and Q. Montardy, Shengzhen, China).
23. October 3-14, 2021, NeuroBridge2021 Summer School in Computational Neuroscience
24. (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA). Cancelled
25. November 2020-, Launched as the World Wide Theoretical Neuroscience Seminar (WWTNS) in November 2020 and renamed in homage to Carl van Vreeswijk in Memoriam (April 20, 2022), Speakers have the occasion to talk about theoretical aspects of their work which cannot be discussed in a setting where the majority of the audience consists of experimentalists. The seminars, held on Wednesdays at 11 am ET, are 45-50 min long followed by a discussion. The talks are recorded with authorization of the speaker and are available to everybody on our YouTube channel.
26. 2022-2023 Online Seminar series ADDICTS (in French). Conferences-addicts.fr.
27. September 4-15, 2022: NeuroBridge2022 Summer School in Computational Neuroscience Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA). *NeuroBridges provides a comprehensive overview of experimental and theoretical approaches to decision-making, covering mechanisms from neural circuits to behavior. In addition to scientific training, the program promotes international collaboration. By bringing together researchers from Mediterranean and Middle Eastern countries, it encourages personal relationships that foster cross-cultural understanding.*
28. June 5-8, 2023, Jerusalem.
29. September 3-14, 2023: NeuroBridge2023 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA).
30. September 16-25, 2024: NeuroBridge2024 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA, Levenes, CNRS).
31. September 7-18, 2025: NeuroBridge2024 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA, Levenes, CNRS).

32. October 4-14, 2026: NeuroBridge2024 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA, Levenes, CNRS).

Ad-hoc reviewer

Nature, Physical Review X & E, PLOS Computational Biology, Physical Review Letters, Elife, Journal of Neuroscience, Journal of Computational Neuroscience, Neural Computation, Neuroscience, Scientific Reports, Frontiers in Comp. Neurosci.

Publications (Full list, updated October 6, 2024)

A. Publications in peer reviewed international journals

1. D. Hansel, F. Bouchet, R. Pellat & A. Ramani, New *theoretical and numerical results in nonlinear gravitational collapse*, Phys. Rev. Lett. 55:487, 1985.
2. A. Georges, D. Hansel & P. Le Doussal, *Exact properties of spin glasses: I) 2D supersymmetry and Nishimori's result,* J. de Phys. (Paris), 46:1827, 1985.
3. A. Georges, D. Hansel, P. Le Doussal & J.-P. Bouchaud, *Exact properties of spin glasses: II) Nishimori line: new results and physical properties*, J. de Physique (Paris), 46:1827, 1985.
4. A. Georges, D. Hansel, P. Le Doussal, J.-M. Maillard & J.-P. Bouchaud, *Rigorous bounds for 2D disordered Ising models*, J. de Phys. (Paris), 47:947, 1986.
5. D. Hansel, F. Bouchet, R. Pellat & A. Ramani, *On the closure of the hierarchy of galaxy correlation function in phase space*, Ap. J., 310:23, 1986.
6. A. Georges, D. Hansel, P. Le Doussal, J.-M. Maillard & J.-P. Bouchaud, Rigorous *bounds and approximations for products of random matrices: exact scaling behavior*, J. Phys. A: Math. Gen., 19: L1145, 1986.
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B.Submitted or in preparation

89. C. van Vreeswijk, G. Mato, D. Barbera, J.J. Pattadkal, N.J. Priebe* and D. Hansel*. The computational power of randomness: *An analysis of selectivity in the thalamic input to mouse visual cortex*, submitted.
90. A. Berrou, M. Harpaz, M Jankowski; D. Hansel, and I. Nelken. *The anatomical mixing of responsive and non-responsive neurons in the ascending auditory pathway,* in preparation
91. A. Berrou, M. Harpaz, M Jankowski, D. Hansel, and I. Nelken. *The emergence of heterogeneous neural responses in the ascending auditory pathway,* in preparation.
92. A. Berrou, R. Darshan, G. Mato, D. Hansel and C. van Vreeswijk. *Theory of orientation selectivity in random network models of the rodent primary visual cortex*, in preparation.

C. Books or book chapters

1. A. Georges, D. Hansel & L. Michel, *Problèmes corrigés de Physique et Chimie posés aux concours d'entrée des Ecoles Normales Supérieures*, Editions Ellipses, 1982.
2. D. Hansel & H. Sompolinsky, Modeling *feature selectivity in local cortical circuits*. In Methods in Neuronal Modeling. From Synapses to Networks, C. Koch and I. Segev (Eds), second edition. MIT Press, Cambridge, MA., 1998.
3. D. Golomb, D. Hansel & G. Mato, Theory *of synchrony of neuronal activity*; In Handbook of Biological Physics, S. Gielen and F. Moss eds, Elsevier, 2001.
4. C. Chow, B. Gutkin, D. Hansel & C. Meunier (Eds.), *Models and Methods in Neurophysics*, Les Houches Summer School, Session LXXX, 2005, Elsevier.
5. A. Leblois, T. Boraud, H. Bergman, & D. Hansel, *Competition between feedback loop as the core of a computational model of the basal ganglia.* In: Recent breakthroughs in basal ganglia research. Bezard, E. (eds), Nova Publishing. New York ,2006
6. D. Battaglia & D. Hansel, *Chaos in networks with delayed local inhibition.* In: Proceedings of the 2nd UniNet Workshop. Kirkilionis, K., Kummer, U. and Stoleriu, I. (eds). Logos Verlag Berlin. pp. 98-104, 2006.
7. D. Hansel, B. Pfeuty,G. Mato, *Phase response curves and dynamical properties of neuronal systems coupled with gap junction*s, in Phase Response Curves in Neuroscience; Theory, Experiment, Analysis, Eds. A. Prinz and N. Schultheiss, Springer, 2012

Invited talks (Since 1999, Updated February 6, 2026)

1. 3-28 May 1999: ICTP School "Neural Information processing" (Trieste, Italy). *Circuitry of the visual cortex and mechanisms for orientation selectivity.*
2. December 1999: AFIRST Conference (Aix en Provence, France).

3. February 24-26, 2000: Neural Dynamics Workshop (Gatsby Institute, London).
4. September 11-14, 2000 : NSI2000 (Dinard, France). *Neurophysique de la synchronisation dans les grands réseaux de neurones.*
5. March 17-20, 2001: Neural Information and Coding Workshop; Invited Posters (Big Sky, USA). 1) *The mechanisms for synchrony and for stability of persistent activity in large neuronal networks. 2) How does noise contribute to contrast invariance of orientation tuning?*
6. May 20-24, 2001: SIAM Conference on Application of Dynamical Systems (Snowbird, USA). *Existence and Stability of Persistent States in Large Neuronal Network.*
7. July-December 2001: Dynamics of Neural Networks: from Biophysics to Behavior (ITP, Santa Barbara, USA).
8. September 2-6, 2001: Joint SIAM-EMS conference (Berlin, replaced by C. van Vreeswijk). *The contribution of noise to contrast invariance.*
9. February 27-28, 2002: Annual retreat of the Zlotowski Center for Neuroscience of Beersheva University (Maale Ha Hamisha, Israel). *Synchrony with inhibition and inhibition of synchrony.*
10. April 8-10, 2002: Workshop on Functional representaticv-october-2013.docons, dynamics of neuronal assemblies (Paris). *Inhibition, synchronisation et stabilité de l'activité réverbérant dans les grands systèmes de neurones.*
11. October 7-18, 2002: Workshop on Neuronal Dynamics, (MBI, Ohio State University (USA). *Emergence of Synchrony in Networks of Electrically Coupled neurons: The role on Intrinsic Currents.*
12. November 18-22, 2002: Workshop on System Level Modeling (MBI, Ohio State University (USA). *The roles of feedback loops in the physiology and the physiopathology of the basal ganglia.*
13. May 13-16, 2003: Annual Meeting of the French Society for Neuroscience Symposium on "Subcortical and cortical mechanisms in action preparation" (Rouen France). *A Model for Action Selection in Basal Ganglia: The Role of Feedback Loops.*
14. June 9-13, 2003: Workshop on "Sensory-Motor System" (MBI, Ohio State University, USA). *The roles of feedback loops in the physiology and the physiopacv-october-2013.docthology of the basal-ganglia.*
15. March 15-18, 2004: Workshop in Mathematical Neuroscience (MSRI, Berkeley, USA). *Neuronal Synchrony and the Interplay between Cellular Properties, Electrical and Inhibitory Synapses.*
16. July 21-22, 2004: Symposium "Reduced Models of Neuronal Excitability and Dynamics of Spike-Generation" (Baltimore, USA). *The Combined Effects of Inhibitory and Electrical Synapses in Synchrony.*
17. July 10-14, 2004: FENS conference (Lisbon, Portugal). *Noise-induced transmutation of nonlinearity and contrast invariance of orientation tuning in V1.* (replaced by C. van Vreeswijk).
18. November 25-28, 2004: Oscillations, Chaos and Network Dynamics in Non-linear Science; international conference in honor of Prof. Kuramoto (Kyoto, Japan). *The role of delays in shaping spatiotemporal patterns of activity in local cortical circuit.*
19. May 22-26, 2005: Symposium on "Reduced neuronal models and complex neurodynamic". SIAM conference on applications of dynamical systems (Snowbird, Utah, USA). *Patterns of activity in neural tissues with spatially modulated connectivity.*

20. July 13-17, 2005: Jacques-Monod conference *Synaptic communication in neuronal networks: from molecules to neural code* (Roscoff, France).
21. February 19-23, 2006: The 11th Otto Loewi Conference: *the Rhythmic Brain* (Eilat, Israel), *Competition between feedback loops underlie normal and pathological dynamics in a model of the basal ganglia.*
22. May 15-19, 2006: *Stochastic and Chaotic Dynamics in Action: from Laser to Brain, from Communication to Medicine* (Yalta, Crimea, Ukraine), *A generic mechanism for synchronized chaos in neuronal networks?*
23. September 1-4, 2006: Mathematical Neuroscience Conference-NeuroMath06 (Andorra) The *ring model: delayed insights.*
24. September 25-29, 2006: Dynamics Days Europe 2006 (Heraklion, Greece) A *generic mechanism for synchronized chaos in neuronal networks?*
25. March 12-14, 2007: Israeli-Italian Bi-National Conference on Brain functions: from Basic Research to Clinical Applications (Weizmann Institute, Rehovot, Israel) *Ionic mechanisms underlying firing patterns in fast-spiking cortical interneurons.*
26. September 16-19, 2007: Workshop on mathematical neuroscience (Montreal, Canada). Competition between *feedback loops underlie normal and pathological dynamics in basal ganglia.*
27. May 19-30, 2008: Network synchronization: from dynamical systems to neuroscience (Leiden, the Netherland) Talk 1: *The ring model of cortical dynamics: an overview*. Talk 2: *Interactions between gamma oscillators and the temporal decorrelation of evoked sensory responses.*
28. July 16-18, 2008: Mathematical Modeling in Neuroscience *(*Kiev, Ukraine). *Balanced Working Memory.*
29. August 11-15, 2008: European Course in Computational Neuroscience (Freiburg, Germany). Talk 1: *The ring model of cortical dynamics: an overview*. Talk 2: *Models of spatial working memory.*
30. February 15-16, 2009: Belfer symposium on neuromechanics and decision making (Technion of Haifa, Israel). *Why the initial behavior of monkey learning operant conditioning tasks is very diverse.*
31. May 26, 2009 : Club des ganglions de la base (Bordeaux, France). *Oscillations: physio pathological role?*
32. July 22, 2009: Workshop on Modern Mathematical Neurodynamic: Bridging Single Cells to Networks, CNS conference (Berlin, Germany).
33. July 23, 2009: Workshop on Cortical oscillations, CNS conference (Berlin, Germany). *Interacting Gamma oscillators.*
34. September 22-23, 2009: Minerva-Weizmann Symposium in Neuroscience *(*Rehovot, Israel),. *Interacting Gamma oscillators.*
35. January 11-13, 2010: Status seminar of the program *Neuroscience and Robotics, - I*sraeli French High Council for Science and Technology Research (Paris).
36. January 11-29, 2010: Summer School in Computational Neuroscience (Valparaiso, Chile).
37. January 31-February 4, 2010: Brain circuits: from receptors to network dynamics (Ein Gedi, Israel). *Balanced spatial working memory.*
38. May 25-28, 2010: American Institute of Mathematical Science (AIMS) Conference on Dynamical System and Differential Equations, Special session: Dynamical Networks and their applications in Neuroscience (Dresden, RFA).

39. June 7-10, 2010: EPSRC Workshop on: Dendrites, Neurons and Networks (Warwick, UK). *Network Mechanisms of Visuospatial Working Memory.*
40. June 11, 2010: Symposium Physiology of Basal Ganglia- Homage to Jean-Michel Deniau (College de France, Paris). *Competition between Feedback Loops Underlies Normal and Pathological Dynamics in the Basal Ganglia.*
41. September 27- November 19, 2010: Emerging techniques in neuroscience (Kavli Institute for Theoretical Physics in Santa Barbara). *Balanced working memory.*
42. March 23-25, 2011: USTC-CNRS Joint Symposium on Synaptic Plasticity and Network Dynamics in the Central Nervous System (University of Science and Technology of China, Heifei, China). *The role of short-term plasticity in working memory.*
43. May 31, 2011: Computational Theory Day at the Interdisciplinary Center for Neural Computation, (The Hebrew University of Jerusalem, Israel). *The generation of feature selective activity in balanced cortical networks.*
44. October 3-7, 2011: Mean-field methods and multiscale analysis of neuronal populations (CIRM, Marseille, France). *Mechanism of orientation selectivity in primary visual cortex without orientation map.*
45. January 12-13, 2012: Symposium on Neural Dynamics and Plasticity: from Synapse to Network (Kyoto University, Japan). *The mechanism of orientation selectivity in primary visual cortex without functional map.*
46. May 10-11, 2012: Second symposium on biology of decision making (Paris, France). *Sensorimotor adaptation with rewards* (cancelled).
47. May 20-27, 2012: 7th Crimean School-Workshop on Emergent Dynamics in Oscillatory Networks (Mellas, Crimea, Ukraine), The m*echanism of orientation selectivity in primary visual cortex without orientation map (*cancelled).
48. September 2-7, 2012: Neural Coding, 2012, (Prague, Czech Republic). *The mechanism of orientation selectivity in primary visual cortex without orientation map.*
49. May 19-21, 2013: Workshop on Computational Neuroscience, (NYU-Abu Dhabi, EAU). *The balanced state: the standard model and beyond; Public* Lecture: *The power of theory, From Physics to Neuroscience.*
50. May 29-30, 2013: Symposium on Biology of Decision-Making (Paris). I*nterference and shaping in sensorimotor adaptations with rewards.*
51. June 9-15,2013: Workshop on Dynamics of cortical and cortical-subcortical circuits, Janelia Farm, USA. *The mechanism of orientation selectivity in primary visual cortex without map.*
52. July 3-4, 2013: Neural Coding: Information beyond Shannon 2013, Prag, Czech Republic. *(cancelled)*
53. July 2-19, 2013: *IAS* Program in Statistical Physics and Computational Neuroscience, Hong Kong. First talk: *The balance state: the standard model and beyond;* Second talk*: The mechanism of orientation selectivity in primary visual cortex without map.*
54. July 17-19, 2013: StatPhysHK Conference. Statistical Physics and Computational Neuroscience, Hong Kong.
55. October 30, 2013. Workshop on modeling in neurobiology, College de France, Paris. *The balance state: the standard model and beyond.*
56. January 20-24, 2014: LACONEU Summer school, Valparaiso, Chile. Talk 1: *The balanced state: the standard model and beyond.* Talk 2: *The mechanism of orientation selectivity in rodents.*

57. February 11-14, 2014: Fifth France-Israel Binational Conference in Neuroscience, Sde Boker, Israel.
58. February 26-27, 2014: Workshop on cortical dynamics: from the architecture to functions, (University of Kyoto, Japan). *The balanced state: the standard model and beyond.*
59. June 10-13, 2014: Sino-French Conference in Computational Neuroscience (Jiao Tong University, Shanghai, China). *Chaotic Rate Dynamics in Large Neuronal Networks.*
60. October 21-23, 2014: Animal vocalizations: Songs in Jerusalem (Jerusalem, Israel). *A network mechanism accounting for universality in babbling vocalization.*
61. December 1-3, 2014: NEUROFrames: pre-kick off meeting of the International Research Network in Neurosciences France- South America (Buenos Aires, Argentina).
62. February 9-12, 2015: Workshop on Computational Neuroscience, (NYU-Abu Dhabi, EAU). *A network mechanism accounting for universality in babbling vocalization.*
63. May 26-27, 2015: Annual retreat of the Department of Neuroscience of Pasteur Institute. (Paris). *The Balanced State: From Physics to Function.*
64. June 8-10, 2015: Meeting in honor of David McLaughlin, on his seventieth birthday (NYU Shanghai, Shanghai, China*). Central network mechanisms of babbling behavior.*
65. July 22-23, 2015: Workshop on oscillations, 2015 CNS Conference (Prague, Czech Republic, replaced by T. Arakaki).
66. July 27-31, 2015: Dynamics of Coupled Oscillators: 40 years of the Kuramoto Model (DYCOSC15), (Max Planck Institute for the Physics of Complex Systems, Dresden, Germany, canceled).
67. 2-12, September 2015: 13th Advanced Summer Course on Computational Neuroscience, Göttingen, *Rate chaos in neuronal networks.*
68. February 9, 2016*: COSYNE2016, Snowbird, USA. Workshop: Coding, correlations, and the dimensionality of neural activity. Random feedforward connections account for orientation selectivity of the thalamic input in rodent visual cortex.*
69. March 1, 2016: COSYNE2016, Snowbird, USA. Workshop: Recurrent spiking neural networks - dynamics, learning, computation. *On spikes, rates and chaos.*
70. May 19-20, 2016: The Early Visual System: comprehensive and data-driven modeling, European Institute for Theoretical Neuroscience, Paris, France. *Theory of orientation selectivity in mouse primary visual cortex.*
71. June 24-25, 2016: Neuropsychiatric disorders -from basic systems to clinical applications: Czech and Slovak Neuroscience Alumni Workshop (Institute of Physiology, the Academy of Sciences Campus, Prague, Czech Republic). *Theory of orientation selectivity without map.*
72. July 10-15, 2016: Sixth French-Israel Neurosciences Conference, (CIRM, Marseille, France). *Balanced or not balanced, what is the question?*
73. December 6-8, 2016: Finding the right balance, (Mitzpe Ramon, Israel). *Functional selectivity in random cortical networks: the role of recurrent connections.*
74. *May 16-18, 2017:* Stochastic Dynamics Out of Equilibrium (Institut Henri Poincaré, Paris). *The correlated cortex.*
75. *June 12-15, 2017:* 6th Conference of the Mediterranean Neuroscience Society (MNS, St Julian, Malta).
76. July 12-13, 2017: 1st A*Midex CompNeuro Days, Workshop on Computational Neuroscience in Marseille (France). *Mice are not Cats: An (almost) exact proof* (cancelled).

77. December *4*-6, 2017: Computational properties of basal ganglia neurons international research workshop of the Israel Science Foundation UK-Israel SYNERGY meeting. (Ramat Gan, Israel). *Recurrent dynamics and competition in basal ganglia.*
78. July 20, 2018: Balance or not balance: what is the question? Colloquium of the Institute of Natural Sciences, (Shanghai Jiao Tong University, Shanghai, China).
79. July 23-27, 2018: International Conference on Applied Mathematics and Computational Neuroscience, In Memory of David Cai *(*Shanghai Jiao Tong University, Shanghai, China).
80. August 6-9, 2018: SIAM Conference on the Life Sciences on Information Processing in Neuronal Networks. *Mice are not Cats: An (almost) exact proof. (*Minneapolis, USA).
81. August-September 2018. *BRAIN18:* Recording, analyzing, manipulating, and modeling whole brain activity, (UC Santa Barbara, Kavli Institute for Theoretical Physics, Santa Barbara USA). *Emergent of orientation selectivity from random networks in mouse primary visual cortex.*
82. October 12, 2018: INC Day, *Paris Descartes University (Paris). Mice are not cats: An (almost) exact proof.*
83. April 28-May 2, 2019: IBAG2019, Biarritz (France). *Dynamic properties of the cortex-BG loop.*
84. September 2-6, 2019: French German WE-Heraeus-Seminar / Novel Physics in Living Systems? (Roscoff, France) *Mechanisms underlying the response of mouse cortical networks to optogenetic manipulation.*
85. November 3-6, 2019: The Operating Regime of Neural Circuits as a Determinant for Computations (HHMI Janelia Research Campus, Ashburn, USA), *Theory of orientation selectivity in the rodent primary visual cortex.*
86. January 27-31, 2020: Simons Workshop: Physics of neural circuits and network dynamics. (Stony Brook University, USA), *The power of randomness or why mice are not cats.*
87. April 30, 2020: ELSC Weekly Seminar. *Theory of orientation selectivity in mouse primary visual cortex.* The Hebrew University of Jerusalem, Israel. Postponed to 2021.
88. May 11-13, 2020: China-France Workshop: Multiscale dynamics and structure in the brain: Experimental, Computational and Theoretical Challenges *The power of randomness in early vision.* Online.
89. August 23-27, 2021: Dynamics Days Europe 2021, Nice, France *A function for the balanced state: emergence of selective neural responses*
90. January 17-23, 2022: 11th computational neuroscience winter school in Shanghai Jiao Tong University (China), *Theory of orientation selectivity in mouse primary visual cortex.* Online
91. 13-14, September 2022: Bernstein Conference on Computational Neuroscience, Symposium "Major transition in cortical circuit evolution" (Berlin). *The statistics of the thalamo-cortical excitation in mouse V1 layer 4.*
92. April 23-29, 2023: Workshop on Mathematical foundations of biological organization, Mathematisches Forschunsinstiturt Oberwolfach (RFA). The *thalamo-cortical projections in mouse V1: theory vs. experiments. (Declined for personal reasons)*
93. March 8, 2024: Hong-Kong University of Science and Technology, Hong-Kong. *The power of randomness.*
94. March 19, 2024: Jia tong University, Shanghai, China. *Emergent feature selectivity from random networks in rodent visual cortex.*
95. March 21, 2024: New York University, Shanghai, China. *Thalamocortical receptive fields in rodent: Theory vs. experiments.*

96. April 12, 2024: Kleinfeld lab, UCSD, online. *Receptive fields of thalamo-cortical excitation in rodent visual cortex (part1).*
97. May 5, 2024: Kleinfeld lab, UCSD, online. *Receptive fields of thalamo-cortical excitation in rodent visual cortex (part2).*
98. August 4, 2024: Emilio Segre Distinguished Lectures in Physics. Tel Aviv University, Israel. *Physics of the Brains: Between Diversity and Universality.* Postponed.
99. June, 2024: Emilio Segre Distinguished Lectures in Physics. Tel Aviv University, Israel. *Physics of the Brains: Between Diversity and Universality.* Postponed.
100. June 15, 2025: Emilio Segre Distinguished Lectures in Physics. Tel Aviv University, Israel. *Physics of the Brains: Between Diversity and Universality.* Postponed because of the '12 days War'.
101. July 1-3, 2026: Symposium in honor of Yves Frégnac, 'Feature selectivity in random cortical networks: Theory and prediction', Paris.

Grants (Full list since 1999)

2000-2003 PICS CNRS-MAE. Neurophysics and physiology of the motor system; cooperation between Laboratoire de Neurophysique et Physiologie du Système Moteur and the Interdisciplinary Center for Neural Computation. 310 kFF.

2001-2003 ACI-CNRS *Dynamics, selectivity and learning in the extrapyramidal loop: The roles of feedback. (*680 kFF).

2001-2003 CLG-NATO *Spatiotemporal Dynamics of Cortical Networks with Electrical and Chemical Coupling.* 250 kFF.

2002-2004 ACI-MNRT. *The role of electrical synapses in the collective dynamics in the neocortex*. 500 kFF.

2004-2007 ACI-MNRT *Functions and dysfunctions of the basal ganglia network: a new model and its confrontation with experiments*, with Thomas Boraud (Bordeaux). 106 k€.

2007-2010 ACI-MNRT *Rôle fonctionnel des connexions en feedback dans le cortex visuel : une étude combinant approches théoriques et expérimentale*", with Lionel Nowak (Toulouse). 116 k€ for my team.

2008-2010 High Council for Scientific and Technological cooperation between France and Israel, research network program in neuroscience and robotics. *Neural Basis of Decision Making and Reinforcement Learning in the Basal Ganglia. 40 k€ for my team.*

2010-2011 Programme Interdisciplinaire CNRS Neurosciences et neuroinformatique computationnelle Neuro-IC. *The role of striatal feedforward inhibition in the control of absence epilepsy seizures.* 34 k€.

2010-2013 ANR-Complex systems. *A new paradigm form working memory: balanced of excitation/inhibition and synaptic nonlinearities*. 500 k€.

2011 France-China Xu Guangqi cooperation program. *Synaptic plasticity and network dynamics in the central nervous system*, with Guoqiang Bi (USTC, China). 10 k€ for my team.

2011-2012 CNRS/JSPS Cooperation Agreement. *Constructing a new realistic network model of working memory which relies on synaptic nonlinearities*, with Shintaro Funahashi, Kyoto University. 28 k€ for my team.

2012-2014 France-Israel High Council for Scientific Research cooperation agreement. *Is there a value in reinforcement learning* with Yonatan Loewenstein, The Hebrew University. 40 k€ (total).

2012 Paris Descartes-University of Ottawa Collaborative Neuroscience Research Initiative. *Non-stationary cortical balanced states*, with André Longtin, Ottawa. 20 € (total).

2013-2014 CNRS/JSPS Cooperation Agreement. *The Geometry of the Recent Past: Looking for Attractors in Pre-Frontal Cortex Neuronal Dynamics During Working Memory Tasks*., with Shintaro Funahashi, Kyoto University. 20 k€ for my team.

2014 PEPS-PTI (Interdisciplinary programs, CNRS). *Physics of the balance state in cortex.* 4 k€.

2014-2017 ANR (generic). *How the balance of excitation and inhibition shapes orientation selectivity and center-surround interactions in the superficial layers of area V1: a combined experimental and modeling project* (with Lionel Nowak and Frederic Chavane). 512 k€.

2015-2018 ANR-US-French Collaboration (CRCNS/NIH/NSF/ANR). *The balance of excitation and inhibition in sensory cortex* (with Nicholas Priebe and Carl van Vreeswijk). 309 k€ (for my team)

2015-2018 Human Frontier Science Program. *Deciphering brain oxytocin circuits controlling social behavior*, with Shlomo Wagner, (Haifa), Valery Grinvich (Heidelberg) and Joseph Buxbaum (New York). 350 k$ for my team.

2015-2020 HHMI-Janelia Farm. *Visiting Scientists Program*. 15 k$/year (approx).

2017-2020 ANR-US-French-Israel Collaboration (CRCNS/NSF/ANR/BSF). *Processing of complex sounds: Cortical Network Mechanisms and Computations.* With Israel Nelken (Jerusalem, Israel) and Tatyana Sharpee (San Diego, USA). 380 k€ for my team.

2017-2022 International Research Network (IRN, former GDRI): *France-Israel Center for Neural Computation.* Co-directed with Israel Nelken (Jerusalem., Israel). 40 k€/year (approx).

2023-2027 ANR-US-French Collaboration (CRCNS/ANR/DOE). *A combined computational and experimental investigation of the cellular and network basis of visual recognition*, with Mark Bear (MIT), 434 k€ for my team

2024-2028 ANR-US-French-Israel Collaboration (CRCNS/ANR/BSF/NIH). *Thalamocortical vs. recurrent connectivity in active sensation of vibrissa touch*, with David Golomb (Ben Gurion University, Israel) and David Kleinfeld (University of California, San Diego, USA), 330 k€ (approx.) for my team

Organization of scientific meetings (Full list, updated October 6, 2024)

1. March 8, 1997: Symposium: *Synchrony in neural systems: can theoreticians contribute* (with C. van Vreeswijk, Big Sky, USA).
2. February 7-12, 1999: Workshop: *Neurophysics and physiology of motor system* (coordinator, with D. Golomb and C. Meunier, Les Houches, France).
3. July 9-16, 2000: Workshop: *Selectivity of neurons in sensory and motor cortices static and dynamic properties in cortex* (coordinator; with I. Segev and C. Meunier, Paris).
4. July 28- August 29, 2003, Summer School (5 weeks), *Neurophysics and theoretical methods in system neuroscience* (coordinator; with C. Chow, J. Dalibard, B. Gutkin and C. Meunier, Les Houches, France).
5. November 15-18,2004: France-Israel binational conference. *High Brain Functions: Multidisciplinary approach for distributed Neural Systems* (coordinator; with A. Desmedt, T. Boraud, Y. Dudai, Maale Hamisha, Israel).
6. July 2-4, 2007: Second France-Israel binational conference. *Neuromodulation in High Brain Functions* (coordinator, with T. Boraud, A. Desmedt, I. Nelken, B. Bioulac. Bordeaux, France).
7. September 18-19, 2008: *Working memory in Jerusalem*. Satellite Workshop of the 5th Conference of Complexity Science (with Misha Tsodyks, Jerusalem, Israel).
8. June 9-10, 2009: *The Biology of Decision Making* (with T. Boraud and S. Ahmed, Bordeaux, France).
9. December 14, 2009: The 7th Paris Area Computational Neuroscience Day. *Synaptic plasticity: From receptors to networks* (with N. Brunel and D. Holcman, Paris).
10. February 14-18, 2010: Third France-Israel binational conference in Neuroscience, neurology and psychiatry (with G. Richter Levin, M. Maroun and A. Desmedt, Haifa).
11. March 22-24, 2010: Workshop. *Working Memory and Executive control* (with M. Tsodyks, Rehovot, Israel).
12. March 8-9, 2011: International symposium: *Working memory and decision making* (with G. Mongillo, Paris).
13. March 23-25, 2011: USTC-CNRS Joint Symposium. *Synaptic Plasticity and Network Dynamics in the Central Nervous System* (with G. Bi, Heifei, China).
14. December 14-15, 2011: International workshop. *Physiological mechanisms underlying of song learning and production* (with A. Leblois and D. Perkel, Université Paris Descartes).
15. March 25-29, 2012: Fourth France-Israel binational conference in Neuroscience, neurology and psychiatry (with D. Shulz, I. Nelken and A. Desmedt, Aussois, France).
16. December 3-6, 2012: French Chinese Joint Symposium. *Synaptic Plasticity and Network Dynamics in the Central Nervous System* (with G. Bi, Université Paris Descartes).
17. June 10-14, 2013: International conference. *Dynamics of cortical and cortical-subcortical circuits: Interaction between theory and experiments* (with D. Golomb, M. Chklovski, K. Svoboda and J. Dudman, Janelia Farm Research Campus, USA).
18. January 8, 2014: Symposia Series Open questions in Neuroscience- Symposium #1: *Balance of excitation and inhibition in sensory cortex* (coordinator, with F. Chavane and L. Nowak, Université Paris Descartes).
19. March 11, 2014: Symposia Series Open questions in Neuroscience- Symposium #2: *Dynamics of large networks, random matrices and neuroscience* (coordinator, with G. Mongillo and C. van Vreeswijk).

20. May 13, 2014: Symposia Series Open questions in Neuroscience- Symposium #3: New Insights in the Mechanism of Absence Epilepsy (with A. Leblois, Université Paris Descartes).
21. December 9, 2014: Symposia Series Open questions in Neuroscience- Symposium #4: Plasticity and memory: beyond synapses (with C. Levenes and G. Mongillo, Université Paris Descartes).
22. July 11-15, 2016: Sixth France-Israel binational conference in Neuroscience, neurology and psychiatry (with F. Chavane and I. Nelken, Marseille, France).
23. Sept 4-14, 2017: NeuroBridges Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA).
24. July 4-6, 2017: Life between Ethics and Science, Toulouse, France (Université de Toulouse Jean-Jaurès, with Flora Bastiani).
25. October 1-12, 2018: NeuroBridge2018 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University and A. El Hady, Princeton, USA).
26. July 16-17, 2019: CNS2019 workshop on New Perspectives on Cortical Dynamics (Barcelona, Spain, with Ni. Priebe and C. Van Vreeswijk).
27. July, 1-3, 2019 : Vie et relation, (Université de Toulouse Jean-Jaurès, Toulouse, France, with Flora Bastiani,).
28. September 8-19, 2019: NeuroBridge2019 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA)
29. September 6-17, 2020: NeuroBridge2020 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA). Cancelled
30. May 11-13, 2020, China-France Workshop: *Multiscale dynamics and structure in the brain*. (Online, With G. Bi and Q. Montardy, Shengzhen, China).
31. October 3-14, 2021: NeuroBridge2021 Summer School in Computational Neuroscience
32. (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA). Cancelled
33. Coordinator of 23 events in 12 countries (international conferences, workshops, public events…) dedicated to Emmanuel Levinas philosophy and person
34. The Work -Wide Theoretical Neuroscience Seminar (www.wwtns.online) a weekly digital seminar series targeting the theoretical neuroscience community. Speakers have the occasion to talk about theoretical aspects of their work which cannot be discussed in a setting where most of the audience consists of experimentalists. The first seminar of the series was on November 4, 2020. The talks are recorded upon authorization by the speaker and posted on the WWTNS YouTube channel.
35. Online Seminar series ADDICTS (in French). Conferences-addicts.fr.
36. September 4-15, 2022: NeuroBridge2022 Summer School in Computational Neuroscience Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA).
37. September 3-14, 2023: NeuroBridge2023 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA).
38. September 16-25, 2024: NeuroBridge2024 Summer School in Computational Neuroscience (Cluny, France, with Y. Loewenstein, Hebrew University, and A. El Hady, Princeton, USA, Levenes, CNRS).